A Context-Aware Real-Time Facial Emotion Recognition System for False Alert Reduction in Security Applications

Wasala Dharmakeerthige Ujali Imalika Samaraweera

IT22258526

Final Report

B.Sc. (Hons) Degree in Information Technology Specialized in Data Science

Department of Computer Science

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DECLARATION

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ABSTRACT

Facial Emotion Recognition (FER) systems have become increasingly prevalent in surveillance and intelligent monitoring environments. While modern deep learning models achieve strong classification performance, their direct deployment in security applications often results in excessive false alerts due to the absence of contextual reasoning. Emotions such as fear may arise in non-threatening scenarios, leading to alarm fatigue and reduced operator trust.

This dissertation presents a context-aware real-time facial emotion recognition system designed to reduce false alerts in security environments. The proposed framework integrates a lightweight custom Convolutional Neural Network (CNN) capable of classifying six emotional states—angry, sad, happy, neutral, fear, and suspicious—with temporal aggregation and decision-layer contextual reasoning.

Unlike conventional FER systems that rely solely on frame-level predictions, the proposed system employs 10-second temporal windows for aggregation, a time-based rule that restricts fear-triggered alerts to nighttime hours (7 PM – 6 AM), and a persistence mechanism requiring multiple consecutive windows before raising alerts. A dual-output architecture separates emotion analytics from security alerts, enabling privacy-conscious deployment. MongoDB is used for data storage.

Experimental evaluation demonstrates approximately 71% classification accuracy and a 55% reduction in false alerts compared to a baseline system lacking contextual reasoning. The framework operates in real time using CPU-based processing, highlighting its suitability for cost-effective deployment.

The results indicate that decision-layer contextual intelligence significantly enhances operational reliability without requiring retraining or architectural modification of the core FER model.

Keywords: Facial Emotion Recognition, Context-Aware, False Alert Reduction, Temporal Aggregation, Security Surveillance, Convolutional Neural Networks, Real-Time Systems, Decision-Layer Intelligence

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1. INTRODUCTION

1.1 Background & Literature Survey

Facial expressions are one of the most powerful non-verbal communication mechanisms in human interaction. Psychological studies suggest that a significant portion of human emotional communication is conveyed visually through facial cues [1]. The ability to automatically interpret facial expressions using computational systems has therefore become a major research area within Computer Vision, and Data Science.

Facial Emotion Recognition (FER) refers to the automated identification of emotional states from facial images or video streams. Early FER systems relied on handcrafted feature extraction techniques such as Local Binary Patterns (LBP), Histogram of Oriented Gradients (HOG), and geometric facial landmark analysis. While these approaches demonstrated moderate performance, their dependency on manually engineered features limited scalability and robustness.

The advent of deep learning, particularly Convolutional Neural Networks (CNNs), revolutionized image-based classification tasks. CNNs enabled end-to-end feature learning, eliminating the need for manual feature engineering. As a result, FER systems achieved substantial improvements in classification accuracy across benchmark datasets. With increasing computational power and availability of large-scale annotated datasets, [2] [3]CNN-based FER models became the dominant paradigm in emotion recognition research.
Beyond academic benchmarks, FER systems have been increasingly integrated into real-world applications, including:

- Human–Computer Interaction systems
- Healthcare monitoring platforms
- Driver fatigue detection
- Educational analytics
- Retail customer behavior analysis
- Security and surveillance systems

Among these domains, surveillance and security monitoring represent a particularly sensitive application area. In such contexts, emotional states like fear, distress, or suspicious expressions are often interpreted as indicators of potential threats or abnormal events. Automated detection of such states can assist security personnel by providing early warnings.

However, real-world deployment reveals a critical challenge: emotional expressions are inherently context-dependent. An expression of fear during a horror movie screening differs significantly from fear during a security incident. Traditional FER systems treat emotional classification outputs as deterministic indicators without contextual interpretation. This limitation leads to operational inefficiencies, particularly in security applications.

In surveillance environments, frequent false alerts reduce trust in automated systems. Alarm fatigue, a well-documented phenomenon in monitoring systems, occurs when operators become desensitized due to excessive false positives. Consequently, the reliability of FER-based security systems [4]becomes more critical than marginal improvements in classification accuracy.

The present research is motivated by this gap between academic performance metrics and practical deployment reliability.

1.2 Evolution of Facial Emotion Recognition

The development of FER systems can be categorized into three major stages:

1.2.1 Handcrafted Feature-Based Methods

Early FER systems extracted geometric and texture-based features from facial regions. Common methods included:

- Active Appearance Models (AAM)
- Facial Action Coding System (FACS)
- Gabor filters
- Local Binary Patterns (LBP)

While these methods achieved moderate performance, they were highly sensitive to lighting, pose variations, and occlusions.

1.2.2 Deep Learning-Based FER

The introduction of deep convolutional neural networks [5] marked a significant advancement. CNNs enabled hierarchical feature extraction, automatically learning discriminative facial representations [6]. Public datasets such as FER2013, CK+, and AffectNet facilitated large-scale training.

- Modern CNN-based FER systems typically consist of:
- Multiple convolutional layers
- Activation functions (ReLU)
- Pooling layers
- Fully connected layers
- Softmax output layers

These systems achieved improved accuracy and generalization capabilities.

1.2.3 Transformer-Based and Hybrid Models

Recent research introduced transformer-based architectures leveraging self-attention mechanisms. Vision Transformers [7] (ViT) model global dependencies within images, potentially improving performance in complex facial expression recognition tasks. However, transformer models generally require:

- Larger datasets
- High computational resources
- GPU-intensive training

Moreover, most research remains focused on improving classification metrics rather than deployment reliability in security contexts [8].

1.3 FER in Security and Surveillance Applications

FER has been applied in surveillance systems for:

- Crowd emotion analysis
- Suspicious behavior detection
- Distress signal identification

- Abnormal event detection

In many implementations, emotional predictions directly trigger alerts. For example:

- Fear → Security Alert
- Angry → Potential Aggression
- Suspicious → Threat

This direct mapping ignores contextual variables such as:

- Time of day
- Environmental setting
- Emotional persistence
- Behavioral continuity

In public spaces, fear expressions may result from entertainment activities, crowded environments, or social interactions [9]. Therefore, blindly mapping fear to threat generates excessive false alerts.

Such operational issues hinder scalability and reduce system credibility.

1.4 Research Gap

A comprehensive examination of existing Facial Emotion Recognition (FER) literature reveals that, despite significant advancements in deep learning architectures, several critical limitations remain when such systems are deployed in real-world surveillance environments. While academic research has largely focused on maximizing classification performance, practical deployment requirements introduce challenges that remain insufficiently addressed. The following research gaps are identified:

1. Accuracy-Centric Evaluation Paradigm

The majority of FER research prioritizes classification accuracy as the primary evaluation metric. Advances in convolutional neural networks, residual architectures, attention mechanisms, and transformer-based models have consistently aimed to optimize benchmark performance on datasets such as FER2013, CK+, and AffectNet. [10]

These studies typically report:

Overall accuracy

Macro and weighted F1-scores

Cross-dataset generalization

While such metrics are essential for evaluating representation learning quality, they do not

fully capture operational performance in real-time monitoring systems.
In surveillance environments, the cost of false positives may exceed the benefit of marginal accuracy gains. A model achieving 90% accuracy on a balanced dataset may still generate excessive false alerts when deployed continuously in dynamic real-world settings. However, most academic studies do not quantify:

- False alert rates
- Alert frequency stability
- Human-operator workload impact
- Deployment-level reliability

Thus, the evaluation paradigm remains largely disconnected from deployment realities.

2. Lack of Contextual Reasoning in Decision Making

Existing FER systems predominantly operate under an assumption of emotional determinism, where predicted emotional states are directly mapped to semantic interpretations. In surveillance contexts, emotions such as fear, anger, or distress are frequently treated as indicators of potential threat.
However, emotions are inherently context-dependent. An expression classified as fear may result from:

- Environmental stimuli (loud noises)
- Entertainment scenarios
- Social interactions
- Playful or exaggerated behavior
- Genuine emergency situations

Without situational awareness, FER systems treat all fear predictions uniformly, leading to semantic misinterpretation. Most current models focus on feature extraction and classification accuracy, but they do not incorporate contextual modifiers such as:

- Time-of-day constraints
- Environmental conditions
- Behavioral continuity
- Persistence duration

Transformer-based architectures, despite capturing global spatial dependencies, do not inherently encode environmental or situational context unless explicitly modeled.
Consequently, improvements in backbone architecture do not resolve the contextual ambiguity problem.

3. Absence of Deployment-Level Reliability Metrics

Another significant gap lies in the limited use of operational metrics in FER research. Benchmark datasets are static and controlled, whereas surveillance environments are continuous and dynamic.

Key deployment metrics rarely reported include:

- False alert reduction percentage
- Alert stability over time
- Real-time processing feasibility on constrained hardware
- Human interpretability and trust measures

The absence of such metrics prevents comprehensive evaluation of system readiness for real-world applications. This disconnect results in FER models that perform well in academic benchmarks but may be unreliable in practice.

4. Lack of Temporal Stability and Persistence Mechanisms

Most FER systems operate at the frame level, generating predictions independently for each input frame. While this approach simplifies model design, it introduces instability due to: Facial micro-expressions , Lighting variations , Occlusions ,Pose changes ,Model prediction noise .

Transient prediction spikes can result in unnecessary alerts if directly interpreted as security events [4] [9].

Although some video-based FER studies incorporate temporal convolution or recurrent networks, these approaches primarily aim to improve classification performance rather than enforce persistence validation for alert generation. In practical surveillance systems, emotional states indicative of threat should persist over time before triggering alerts.

The absence of structured persistence mechanisms results in systems that are sensitive to short-lived anomalies, reducing reliability.

5. Limited Modeling of Domain-Specific Security Categories

Publicly available FER datasets typically include basic emotional categories: Angry , Sad, Happy , Neutral , Fear , Surprise .

However, the concept of a “suspicious” expression—particularly relevant in surveillance contexts—is rarely explicitly modeled. As a result:

- Models are trained on generic emotional taxonomies.
- Security-relevant facial patterns may not be captured effectively.
- Deployment systems rely solely on traditional emotion categories.

This limitation creates a gap between academic emotion recognition and practical security monitoring requirements.

Table 1:Comparative Analysis of Existing FER Systems and the Proposed System

Feature / Criterion	Research A (CNN-Based FER)	Research B (Transformer-Based FER)	Research C (Video-Based FER)	Proposed System
Primary Objective	Improve classification accuracy	Improve representation learning via attention	Improve temporal classification accuracy	Reduce false alerts in security deployment
Backbone Architecture	Deep CNN	Vision Transformer / Hybrid CNN-Transformer	CNN + Temporal Model (LSTM/TCN)	Lightweight Custom CNN
Context-Aware Reasoning	✗ Not implemented	✗ Not implemented	✗ Limited	✓ Time-based contextual logic
Temporal Aggregation for Alert Stability	✗ Frame-level only	✗ Frame-level only	✓ For classification	✓ For decision-layer validation
Persistence-Based Alert Validation	✗	✗	✗	✓ Multi-window requirement
Deployment-Level Metrics (False Alert	✗ Rarely reported	✗ Rarely reported	✗	✓ Explicitly measured (55%

Rate)				reduction)
Domain-Specific "Suspicious" Class	✗ Standard emotion classes only	✗ Standard emotion classes only	✗	✓ Included
Dual-Output Design (Analytics + Security)	✗	✗	✗	✓ Implemented

1.4.1 Limitations of Transformer-Based FER in Addressing Deployment Challenges

Transformer-based architectures have recently demonstrated strong performance in image recognition tasks due to their ability to model long-range dependencies. In FER applications, vision transformers [10]can improve feature representation by attending to salient facial regions.

However, these models primarily address representational power rather than semantic interpretation at the system level. Specifically:

- Transformers improve spatial attention but do not inherently encode contextual rules.
- They do not incorporate time-of-day logic.
- They do not enforce alert persistence constraints.
- They do not separate analytics from security decision outputs.

Therefore, even state-of-the-art transformer backbones may continue to generate false alerts if deployed without contextual intelligence.

1.4.2 Implication for This Research

Based on the identified gaps, this research shifts the focus from model-centric optimization to system-level intelligence. Instead of redesigning or retraining complex architectures, the proposed approach introduces:

- Temporal aggregation mechanisms
- Time-based contextual filtering
- Persistence validation
- Dual-output system design

This reorientation addresses operational reliability without increasing architectural complexity.

Thus, the fundamental gap addressed in this dissertation lies not in feature extraction but in decision-layer intelligence for real-time surveillance deployment.

1.5 Problem Statement

Traditional facial emotion recognition (FER) systems deployed in surveillance environments operate under a classification-centric paradigm. [11] These systems are designed to predict emotional categories from facial images and frequently map specific emotional states directly to security alerts. In many implementations, emotions such as fear or anger are treated as indicators of potential threat, resulting in automated alert generation.
However, this direct mapping mechanism overlooks two critical aspects of real-world surveillance environments:

- Contextual dependency of emotions
- Temporal instability of frame-level predictions [12]

Emotional expressions are inherently ambiguous and heavily influenced by situational factors. For example, an expression classified as fear may arise in diverse contexts including: Entertainment settings (e.g., watching a horror video)

- Social interactions
- Startle responses
- Playful behavior
- Genuine security threats

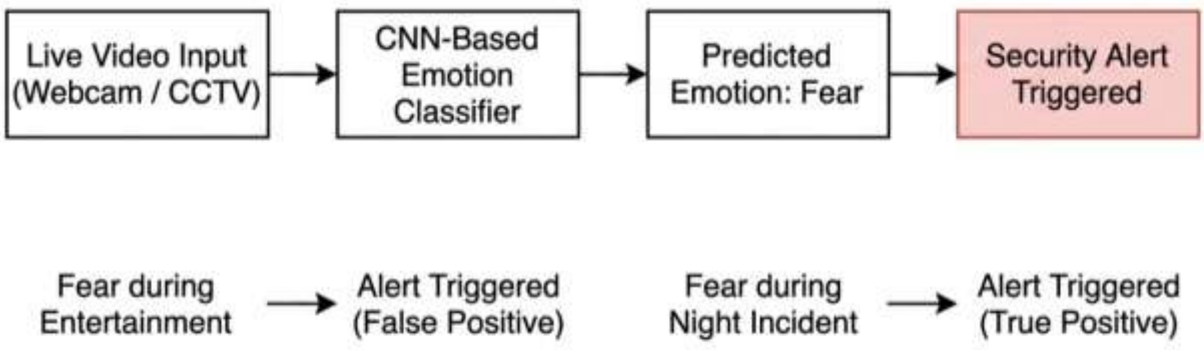

Figure 1:Traditional Facial emotion recognition surveillance pipeline

Without contextual interpretation, FER systems treat all fear predictions uniformly as potential danger signals. This rigid interpretation introduces systemic bias in alert generation.
Furthermore, most FER models operate at the frame level. Each frame is processed independently, and predictions are generated without considering temporal continuity. Due to

natural facial micro-expressions, lighting variations, and pose fluctuations, short-lived prediction spikes frequently occur. When such transient predictions trigger alerts, the system becomes unstable and unreliable.

1.5.1 Consequences of Context-Free FER in Surveillance

The absence of contextual reasoning and temporal validation leads to several operational challenges:

1. Alarm Fatigue

Excessive false alerts desensitize human operators. Alarm fatigue reduces vigilance and increases the probability that genuine threats are overlooked.

2. Reduced Operator Trust

Frequent false positives erode trust in automated systems. Operators may begin ignoring alerts, undermining the system's intended purpose.

3. Inefficient Resource Allocation

Security personnel may respond unnecessarily to benign events, increasing operational cost and reducing efficiency.

4. Limited Deployment Feasibility

Systems that generate unstable alerts cannot be scaled to real-world environments, particularly in institutions with limited monitoring staff.

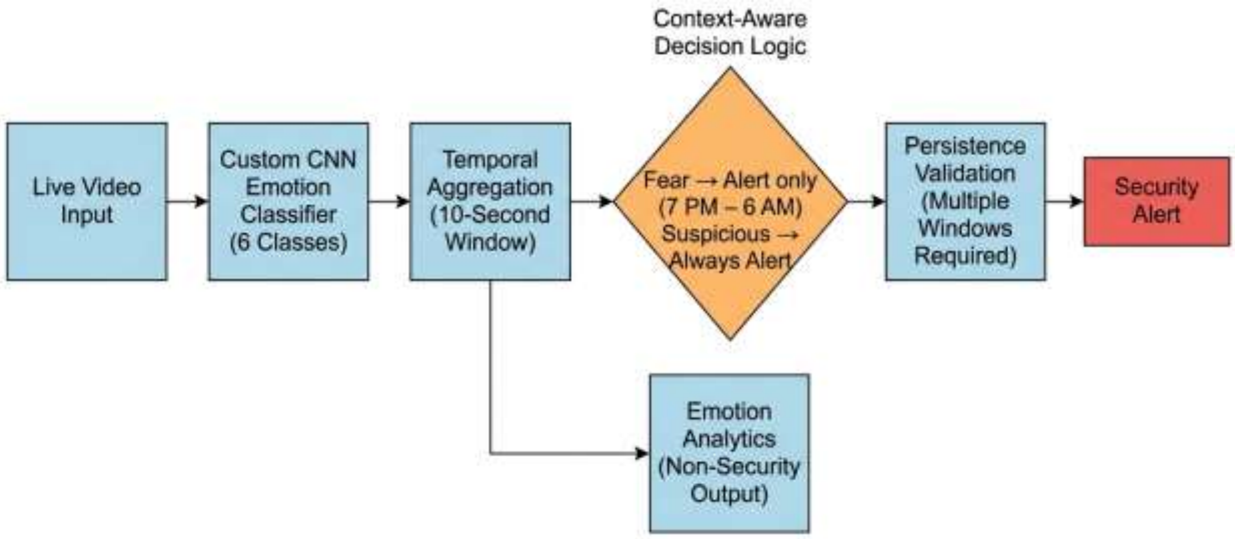

Figure 2:Emotion recognition pipeline with decision layer

1.5.2 Research Question

This research addresses the following central question:

How can a real-time facial emotion recognition system be enhanced with contextual reasoning and temporal validation mechanisms to reduce false alerts in security applications without modifying the core classification architecture?

This question shifts the focus from model accuracy improvement to deployment reliability optimization.

Rather than retraining deep neural networks or designing heavier architectures, the present study investigates whether intelligence can be introduced at the post-classification stage through:

- Temporal aggregation
- Context-aware rule integration
- Persistence-based validation
- Dual-output decision separation

This approach emphasizes architectural intelligence rather than model complexity.

1.6 Research Objectives

1.6.1 Main Objective

To design and implement a context-aware real-time facial emotion recognition system that reduces false alerts in surveillance environments while maintaining real-time performance.

1.6.2 Specific Objectives

1. Develop a lightweight custom CNN [13]capable of classifying six emotion categories: Angry , Sad, Happy , Neutral , Fear ,Suspicious .
2. Implement real-time facial detection and emotion classification using webcam input.
3. Introduce temporal aggregation using 10-second windows to stabilize predictions.
4. Design a time-based contextual rule:

   Fear generates alerts only between 7 PM and 6 AM.
5. Implement a persistence mechanism requiring multiple consecutive windows before raising alerts.
6. Develop a dual-output architecture:

   Emotion analytics stream

   Security alert stream
7. Integrate MongoDB for structured storage of events.
8. Develop a alert interface for real-time visualization.
9. Evaluate system performance using:
   - Accuracy

- Macro F1-score
- Confusion matrix
- False alert reduction rate
- Real-time latency

This chapter introduced the background of facial emotion recognition, its evolution, and its application in surveillance systems. It identified a critical research gap in contextual reasoning and deployment reliability. The problem statement and research objectives were clearly defined, establishing the foundation for the methodological framework presented in the next chapter.

The subsequent chapter details the complete methodology, including CNN design, dataset preparation, temporal aggregation [12], contextual decision logic, system architecture, and implementation strategy.

2. METHODOLOGY

2.1 Requirement Gathering and Analysis

The development of a context-aware real-time facial emotion recognition (FER) system for security applications necessitates a structured and deployment-oriented requirement analysis process. Unlike conventional FER research [14]that primarily emphasizes benchmark accuracy improvements, this study focuses on operational reliability, contextual reasoning, and false alert reduction in real-time surveillance environments.

The requirement gathering process involved:

- Systematic literature review of FER-based surveillance systems [15]
- Analysis of common failure patterns in emotion-driven alert systems
- Identification of operational challenges such as alarm fatigue
- Evaluation of real-time processing constraints
- Assessment of hardware limitations for practical deployment

The core premise guiding requirement analysis was that emotion classification accuracy alone does not guarantee reliable security behavior. Instead, stability over time and contextual validation are critical.

The requirements were categorized into functional and non-functional groups to ensure both behavioral correctness and system robustness.

2.1.1 Functional Requirements

Real-Time Facial Detection

The system must continuously detect and track human faces in live video streams captured through standard webcams or surveillance cameras. Face detection must operate at real-time frame rates without requiring GPU acceleration.

MediaPipe was selected for face detection due to its lightweight implementation and suitability for CPU-based inference.

Multi-Class Emotion Classification

The system must classify facial expressions into six emotion categories:Angry , Sad , Happy, Neutral , Fear , Suspicious .

The inclusion of the suspicious category is domain-specific and designed to model behavioral cues associated with heightened security risk.

Temporal Stabilization

Frame-level predictions are inherently unstable due to:

- Lighting variation
- Occlusions
- Head movement
- Facial micro-expressions

Therefore, predictions must be aggregated across fixed 10-second temporal windows to improve stability.

Context-Aware Security Logic

The system must incorporate contextual reasoning rules:

Fear should trigger alerts only during night hours (19:00–06:00)

Suspicious emotion should always be considered security-relevant

This logic must operate independently of the CNN classifier. [9]

Persistence-Based Validation

Alerts must only be triggered when security-relevant emotion windows [16] persist across multiple consecutive windows. This prevents transient misclassifications from generating false alarms.

Dual Output Architecture

The system must produce two independent outputs:

Emotion analytics output (non-security monitoring)

Security alert output (validated events only)

This separation enhances privacy awareness and system modularity.

Event Logging

All emotion windows and security events must be stored in MongoDB with:Timestamp, Dominant emotion , Confidence score , Window duration and Alert status.

Real-Time Notification Interface

Security alerts must be displayed via a lightweight graphical interface. The interface must operate without reliance on external cloud messaging services.

2.1.2 Non-Functional Requirements

Real-Time Responsiveness
The system must operate under CPU-only constraints with low inference latency.

Deployment Compatibility

The architecture must be backbone-agnostic, allowing replacement of the CNN with transformer models without modifying the decision logic layer.

Reliability

False alerts must be minimized while preserving detection of genuine security events.

Maintainability

Contextual rules must be adjustable without retraining the neural network.

Scalability

The system must support integration with larger surveillance networks.

2.2 Feasibility Study

A comprehensive feasibility assessment was conducted to evaluate the viability of deploying the proposed context-aware FER system in real-world surveillance environments. The feasibility analysis considered three primary dimensions: technical feasibility, operational feasibility, and economic feasibility.

2.2.1 Technical Feasibility

The proposed system was implemented and evaluated using CPU-only processing without GPU acceleration. This hardware configuration represents a mid-range consumer-grade computing platform, commonly available in institutional and commercial settings.
Real-time inference was achieved using optimized libraries such as TensorFlow, OpenCV, and MediaPipe. The lightweight custom CNN architecture was intentionally designed to reduce computational complexity while maintaining acceptable classification performance. The model processes live video frames with stable latency [17], enabling real-time monitoring without performance degradation.
The successful execution under CPU-only conditions confirms that:

- Dedicated GPU infrastructure is not mandatory.
- The system can operate on standard surveillance workstations.

- Edge-device deployment is feasible in resource-constrained environments.

This technical validation demonstrates that the framework is not restricted to laboratory environments and can be integrated into practical surveillance setups without significant hardware upgrades.

2.2.2 Operational Feasibility

Operational feasibility assesses whether the system aligns with standard surveillance workflows and security protocols.

The system architecture integrates seamlessly with conventional monitoring systems by supporting:

- Continuous video acquisition from webcam or CCTV sources.
- Automated emotion detection without human intervention.
- Real-time alert visualization interface.
- Event logging in MongoDB for audit and review.

Unlike intrusive surveillance technologies, the proposed framework does not require active operator input for classification. Instead, it functions as an assistive monitoring layer that enhances situational awareness.

Additionally, the dual-output design ensures that:

- Emotion analytics can be used independently for behavioral insights.
- Security alerts are generated only after contextual validation.

This separation improves usability and reduces unnecessary alert interruptions [9] [16]. The workflow aligns with existing monitoring infrastructure, minimizing training requirements for security personnel.

2.2.3 Economic Feasibility

The economic feasibility of the system is supported by its reliance on widely available and cost-effective components.

The system utilizes:

- Open-source software libraries (TensorFlow, Keras, OpenCV, MediaPipe).
- Python-based implementation without proprietary dependencies.
- MongoDB Community Edition for data storage.
- Standard webcam devices for video input.

No specialized hardware, paid APIs, or enterprise licenses are required for core functionality. This significantly reduces deployment cost compared to GPU-dependent deep learning systems.

The absence of dedicated GPU requirements further lowers operational expenses, making the solution accessible to:

- Small retail businesses
- Educational institutions
- Transportation facilities
- Medium-scale enterprises

From a commercialization perspective, the system supports scalable deployment models, including edge-device installation or centralized monitoring setups.

Overall, the cost-to-performance ratio makes the framework economically sustainable while maintaining technical effectiveness.

The evaluation confirms that the proposed context-aware FER system is technically implementable, operationally compatible with surveillance workflows, and economically viable for real-world deployment. The lightweight architecture, CPU-based execution, and modular design collectively position the framework as a practical and scalable solution for security applications.

2.3 System Design

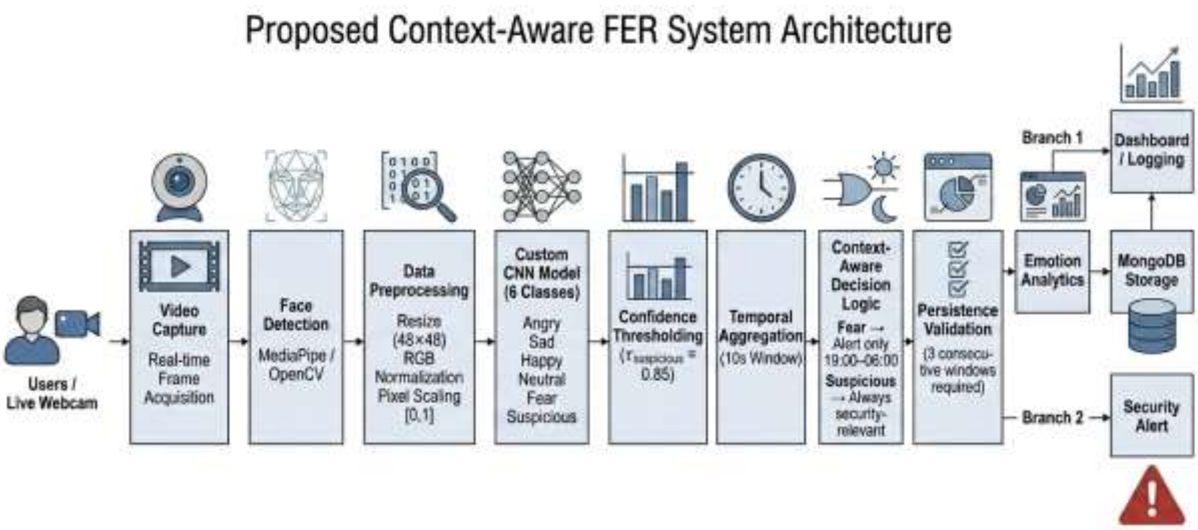

Figure 3:System Architecture Diagram

Figure X illustrates the overall architecture of the proposed context-aware facial emotion recognition (FER) system. The framework follows a real-time sequential processing pipeline beginning with live video capture and face detection using MediaPipe and OpenCV. Detected facial regions undergo preprocessing, including resizing (48×48), RGB normalization, and pixel scaling, before being passed to a custom CNN model that classifies six emotion categories: angry, sad, happy, neutral, fear, and suspicious.

To enhance reliability, confidence thresholding is applied, followed by a 10-second temporal aggregation window to suppress frame-level noise. The aggregated emotion is processed through a context-aware decision layer, where suspicious emotions [9] are always considered security-relevant, while fear triggers alerts only during nighttime (19:00–06:00). A persistence validation mechanism further requires three consecutive relevant windows before alert activation.

The architecture concludes with a dual-output design: emotion analytics are stored in MongoDB for monitoring purposes, while validated events trigger real-time security alerts. This modular design improves deployment-level reliability without modifying the core classification backbone.

2.4 Understanding the Research Domain

Facial Emotion Recognition (FER) is traditionally positioned within the domains of affective computing and computer vision, where the primary objective is to infer emotional states from facial imagery using supervised learning models. Over the past decade, advances in deep learning—particularly Convolutional Neural Networks (CNNs) and transformer-based architectures—have significantly improved classification accuracy on benchmark datasets such as FER2013 and AffectNet.

From a data science perspective, conventional FER research primarily focuses on optimizing predictive performance metrics such as:

- Accuracy
- Precision
- Recall
- F1-score

Cross-dataset generalization

These improvements are achieved through:

- Feature representation learning
- Architectural depth

- Transfer learning
- Attention mechanisms
- Data augmentation

However, such studies are largely evaluated in controlled or semi-controlled benchmarking environments. While high classification accuracy is important, it does not directly translate to operational reliability in real-time surveillance systems.

2.4.1 FER as a Predictive Modeling Problem

In traditional machine learning framing, FER is treated as a multi-class classification problem:

$$\boldsymbol{f}: \boldsymbol{X} \to \boldsymbol{Y}$$

Where:

- **X** represents facial image features
- **Y** represents discrete emotion labels

The goal is to minimize empirical risk:

$$R^{(f)} = \frac{1}{n} \sum_{i=1}^{n} L\,(y_i, f(x_i))$$

Where L is typically cross-entropy loss.

However, in deployment-level surveillance systems, the objective function differs fundamentally. The system must minimize:

- False alert rate
- Alert instability
- Operator fatigue
- Operational inefficiency

This reframes FER not merely as a classification problem, but as a decision-support system under uncertainty.

2.4.2 Surveillance as a Decision Science Problem

This From a data science standpoint, surveillance systems operate within a decision-theoretic

framework rather than a pure prediction framework.
In this setting, the classifier output is not the final objective. Instead, it is an intermediate probabilistic signal that informs a downstream decision function:

$$D: (E, t, H) \rightarrow \{Alert, NoAlert\}$$

Where:
E = Predicted emotion
t = Temporal context
H = Historical emotional stability

The challenge lies in transforming noisy probabilistic outputs into reliable operational decisions.
This introduces concepts from:

- Statistical reliability engineering
- Signal smoothing
- Risk minimization
- Context-aware inference
- Sequential decision modeling

2.4.3 The Ambiguity of Emotional Signals

Human emotions are inherently ambiguous and context-dependent. In real-world environments, the same emotional expression can correspond to multiple semantic interpretations.
For example, fear may represent:

- Amusement in entertainment settings
- Excitement in gaming environments
- Surprise during social interaction
- Genuine distress during threatening situations

From a probabilistic perspective, emotion alone does not uniquely determine threat level. The conditional probability:

$$P(Threat \mid Fear)$$

is not constant and varies depending on situational variables such as time of day, location, and surrounding activity.

Therefore, a direct mapping:

$$\text{Fear} \rightarrow \text{Alert}$$

ignores conditional dependencies and leads to inflated false positives.

2.4.4 Stability and Temporal Consistency

Another critical aspect in real-time systems is temporal stability.

Frame-level predictions exhibit high variance due to:

- Sensor noise
- Illumination changes
- Micro-expressions
- Pose variation

In statistical terms, the prediction sequence $\{E_t\}$ contains short-term noise components that must be filtered to extract stable behavioral patterns.

This aligns with concepts from:

- Time-series smoothing
- Rolling window aggregation
- Majority voting
- Sequential filtering

Without temporal validation, systems become sensitive to outliers, increasing false alert probability.

2.4.5 Deployment-Level Metrics in Data Science

A major gap in FER research lies in the limited use of deployment-centric evaluation metrics.

- Most studies report:
- Classification accuracy
- Cross-validation performance
- ROC curves

However, real-world surveillance systems require evaluation based on:

- False alert rate

- Alert persistence
- Response stability
- Computational efficiency
- Latency

These metrics align more closely with applied data science and systems engineering than pure machine learning research.

2.4.6 From Feature Learning to Reliability Engineering

Transformer-based architectures demonstrate impressive representational capabilities. However, increasing feature extraction capacity does not inherently address contextual misinterpretation.

This research repositions the problem:

Rather than improving feature representation, it introduces decision-layer intelligence to enhance reliability.

The contribution is therefore not:

- A deeper CNN
- A novel attention mechanism
- A new backbone architecture

Instead, it introduces:

- Context-aware probabilistic filtering
- Temporal validation
- Persistence-based event confirmation
- Dual-output analytical separation

2.4.7 Data Science Framing of the Proposed Approach

The proposed framework can be interpreted as a layered modeling approach:

1. **Predictive Layer (CNN)**

Learns emotional representations from facial features.

2. **Statistical Filtering Layer (Temporal Aggregation)**

Reduces variance and suppresses noise.

3. **Contextual Decision Layer**

Applies domain knowledge and situational constraints.

4. **Persistence Validation Layer**

Implements sequential consistency requirements.

This layered architecture aligns with modern data science principles where predictive modeling is complemented by domain-aware post-processing to ensure operational reliability.

2.5 Methodology

The methodological framework integrates supervised deep learning, statistical filtering, and rule-based contextual reasoning into a unified real-time surveillance pipeline. The design philosophy emphasizes deployment reliability rather than solely maximizing classification accuracy.

The complete pipeline consists of:

- Data acquisition
- Data preprocessing
- Model architecture design
- Confidence-based filtering
- Temporal aggregation
- Context-aware decision logic
- Persistence validation

Each stage contributes to reducing false alert generation while maintaining acceptable classification performance.

2.5.1 Data Acquisition

The primary dataset used in this study was the publicly available Sri Lankan Facial Expression Dataset, obtained from IEEE DataPort. [18] This dataset contains labeled facial images representing five standard emotional categories:

- Angry
- Fear
- Happy
- Neutral

- Sad

The total dataset size used in this study comprised 931 facial images distributed as follows:

- Angry: 115
- Fear: 191
- Happy: 204
- Neutral: 318
- Sad: 103
- Suspicious: 114

It is important to clarify that the original IEEE dataset does not contain a “suspicious” emotion class. Therefore, the suspicious category used in this research was synthetically generated and curated separately to support domain-specific security modeling.

Dataset Limitations

While the Sri Lankan dataset enhances regional validity, several limitations must be acknowledged:

Moderate dataset size (931 images total).

Class imbalance across categories.

Synthetic nature of suspicious class. [19]

Limited variability compared to large-scale global datasets.

Despite these limitations, the dataset is suitable for evaluating the primary objective of this research: deployment-level reliability enhancement rather than large-scale benchmark superiority.

Alignment with Research Objective

It is important to emphasize that this study does not claim to redefine emotional taxonomy. Instead, the suspicious category serves as a security-oriented behavioral indicator within a controlled experimental framework.

The dataset design supports the central research question:

How can contextual reasoning and temporal validation reduce false alerts in security systems without modifying the underlying classifier?

Ethical Considerations in Synthetic Data

The introduction of a synthetically generated “suspicious” class requires careful ethical consideration. Unlike standard emotions such as happiness or fear, “suspicious” is not a recognized psychological category but a domain-specific behavioral indicator intended for security-oriented applications. It does not represent an inherent personality trait or diagnostic label.

To minimize ethical risks, all synthetic images were manually reviewed for plausibility, and no real individuals were labeled as “suspicious” without consent. The class definition was restricted strictly to observable facial expression patterns [20] [21], without using demographic attributes such as ethnicity, gender, or age.

It is acknowledged that synthetic data may introduce bias and limit generalizability. Therefore, the suspicious class should be interpreted as a controlled experimental construct rather than a definitive behavioral classifier. Moreover, the system operates as a decision-support tool and does not make autonomous enforcement decisions, thereby reducing risks of misclassification or discrimination.

Future work should incorporate ethically sourced real-world behavioral data, cross-demographic validation, and fairness auditing to further enhance robustness and accountability.

2.5.2 Data Preprocessing

Preprocessing ensures standardized input representation and stable model convergence.

The following steps were applied: [22]

Face Detection

Faces were detected using a lightweight face detection module to isolate relevant regions and remove background noise [23].

Resizing

All facial images were resized to 48 × 48 pixels to balance computational efficiency and spatial feature retention.

RGB Normalization

Color channel normalization ensures consistent input scaling across varying lighting conditions.

Pixel Scaling

Pixel intensities were scaled to the range [0,1] to stabilize gradient updates during training.

One-Hot Encoding

Emotion labels were encoded into categorical vectors for multi-class classification using

Softmax output.

These preprocessing steps reduce variance in raw data and improve generalization [24].

2.5.3 Model Architecture

A lightweight Convolutional Neural Network (CNN) was designed to achieve real-time performance under CPU constraints.

The architecture consists of:

- Convolutional layer (32 filters) + ReLU activation
- Batch normalization for training stability [25]
- Max pooling for spatial reduction
- Convolutional layer (64 filters) + ReLU
- Dropout layer to mitigate overfitting
- Convolutional layer (128 filters) + ReLU
- Flattening layer
- Fully connected dense layer (128 units)
- Softmax output layer (6 classes)

The architecture balances representational capacity with computational efficiency.

Loss Function

Categorical cross-entropy was used:

$$L = -\sum_{i=1}^{C} y_i \log(\hat{y}_i)$$

Where:

- y_i is the true label
- $\hat{y}_i$ is the predicted probability
- C = 6 classes

Optimization

The Adam optimizer was selected for adaptive gradient updates. [26]

Training configuration: Epochs: 15 , Batch size: 32 and Early stopping applied.

Performance Metrics

Fine-tuned test accuracy:

$$Accuracy = 0.7235$$

Macro F1-score:

$F1_{macro} = 0.70$

Suspicious class F1-score:

$$F1_{suspicious} = 0.98$$

The high F1-score for the suspicious class indicates strong precision-recall balance for the most security-critical category.

2.5.4 Confidence Thresholding

To prevent low-confidence predictions from influencing downstream decisions, a confidence filtering mechanism was implemented.

Let; $p_{\max} = \max(\hat{y})$

A prediction is accepted only if: $p_{\max} \geq \tau$

For the suspicious class: $\tau_{suspicious} = 0.85$

This higher threshold prioritizes precision over recall. In security systems, false positives are often more damaging than missed detections due to alarm fatigue. [27]

Confidence filtering therefore acts as a probabilistic reliability gate before temporal aggregation.

2.5.5 Temporal Aggregation

Frame-level predictions are inherently noisy. To reduce prediction variance, a fixed 10-second sliding window is used.

The dominant emotion for a window is computed as:

Where: $E_{window} = arg \max_{e} \sum_{i=1}^{N} I\ (e_i = e)$

e_i represents the frame-level prediction

N is the number of frames in the 10-second interval

I(.) is the indicator function

This majority voting mechanism suppresses transient spikes and stabilizes emotional trends.

Temporal aggregation transforms high-frequency predictions into statistically meaningful segments.

2.5.6 Context-Aware Decision Rule

After temporal stabilization, context-aware filtering is applied.

$$R(E,t) = \begin{cases} 1 & if\ E = suspicious \\ 1 & if\ E = fear \wedge t \in [19{:}00, 06{:}00] \\ 0 & otherwise \end{cases}$$

This rule integrates domain knowledge into the decision layer.

The fear emotion is treated as context-sensitive due to its semantic ambiguity. Restricting fear-based alerts to nighttime reduces false alarms during benign daytime activities.

This logic operates independently of the CNN, ensuring modularity.

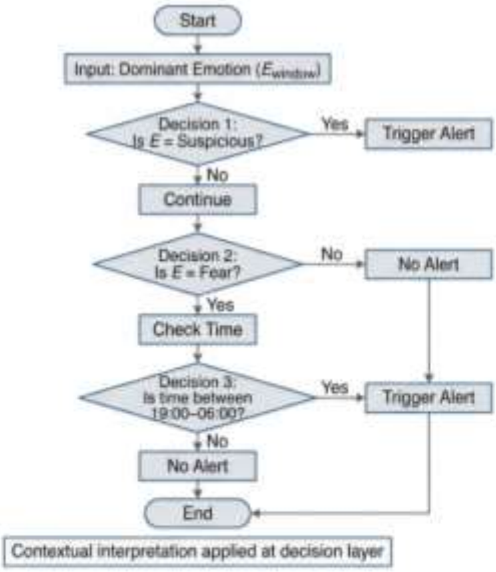

Figure 4:Context-Aware Flowchart

2.5.7 Persistence Validation

To further enhance reliability, a persistence requirement is imposed.

An alert is triggered only if: $\sum_{k=1}^{3} R\ (W_k) \geq 3$

Where:

- W_k represents consecutive emotion windows
- Minimum required windows = 3

Given a 10-second window, the minimum sustained duration is approximately 30 seconds (extending to 60 seconds depending on overlap).

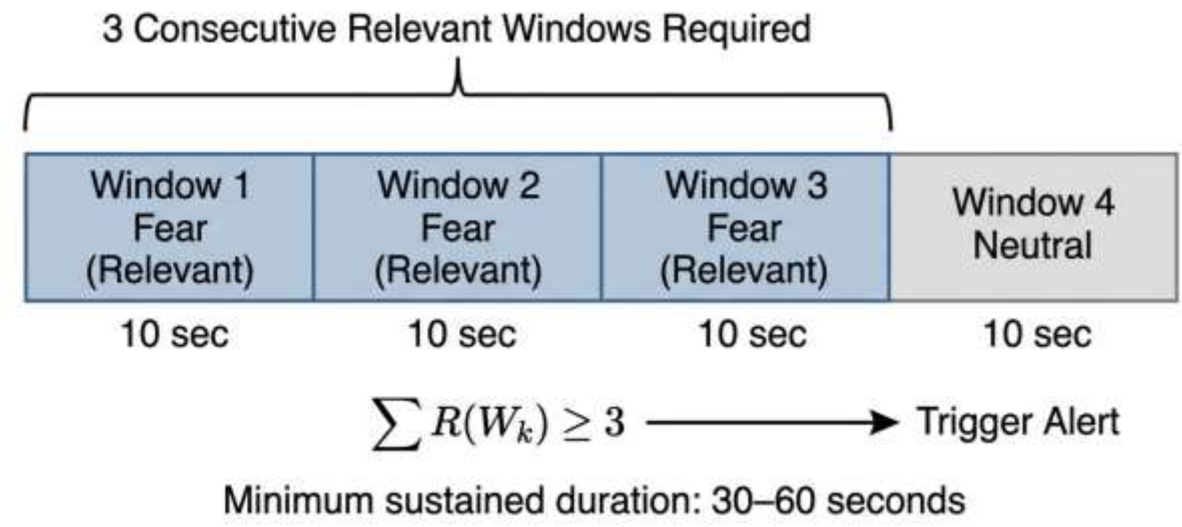

Figure 5:Persistance-Based Alert Validation Mechanism

Persistence validation ensures that only sustained behavioral patterns trigger security alerts.

This sequential consistency mechanism significantly reduces false positives.

2.6 Commercialization Aspects

The proposed system demonstrates strong commercialization potential due to its lightweight and modular architecture.

Application Domains

- The framework can be deployed in:
- Smart retail analytics platforms
- Public transport security systems
- Educational institutions
- Corporate premises
- Smart city monitoring infrastructures

Business Model Possibilities

The system supports multiple monetization strategies:

1. Subscription-Based Analytics
   Businesses can subscribe to emotion analytics dashboards for behavioral insights.

2. Security Alert Licensing
   The context-aware security module can be licensed as an add-on for surveillance systems.
3. Edge Device Deployment
   Compact CPU-based devices can be deployed at edge locations.

Competitive Advantages

- CPU-only implementation reduces hardware cost
- No dependency on cloud inference
- Modular design allows backbone upgrades
- False alert reduction (55%) improves usability
- Dual-output design supports both analytics and security

The low operational cost combined with deployment reliability positions the framework as a commercially viable solution.

2.7 Implementation and Testing

The proposed context-aware FER framework was implemented using a modular and scalable software stack designed to support real-time CPU-based deployment [28]. The system integrates deep learning inference, temporal logic, database storage, and graphical alert visualization within a unified pipeline..

2.7.1 Implementation Stack

The system was developed using the following technologies:

- Python – Core programming language
- TensorFlow / Keras – Deep learning model training and inference
- OpenCV – Image processing and frame handling
- MediaPipe – Real-time face detection [23]
- MongoDB – Event logging and emotion analytics storage

The modular design ensures that each component operates independently, allowing upgrades or replacements (e.g., switching CNN to transformer) without altering the decision logic layer.

2.7.2 Implemented Key Features

The following key functional components were implemented and validated:

Table 2:key functional components

Feature ID	Implemented Feature	Description
F1	Real-Time Face Detection	MediaPipe-based face detection from live webcam stream
F2	Multi-Class Emotion Classification	CNN-based 6-class emotion prediction
F3	Confidence Thresholding	Suspicious class filtered using $\tau = 0.85$
F4	Temporal Aggregation	10-second sliding window majority voting
F5	Context-Aware Logic	Fear alerts restricted to 7 PM – 6 AM
F6	Persistence Validation	Minimum 3 consecutive windows required
F7	Dual Output System	Separate emotion analytics and security alert channels
F8	MongoDB Logging	Window-level emotion storage and event logging

2.7.3 Testing Procedures

System testing was performed at multiple levels to ensure functional correctness, reliability, and operational stability. Unit testing was conducted for individual modules including face detection, CNN inference, temporal aggregation, contextual decision logic, and database logging to verify expected behavior. Scenario-based evaluations were carried out using simulated real-world conditions to assess decision-layer performance under varying emotional and temporal contexts. Day and night simulations were specifically implemented to validate the time-based fear filtering mechanism. A comparative analysis between a baseline emotion-only system and the proposed context-aware framework demonstrated a false alert reduction of approximately 55% $\left(F_p \approx 0.45F_b\right)$. Additionally, latency measurements confirmed stable real-time inference under CPU-only implementation, validating the system’s suitability for live deployment environments.

2.7.4 Test Case Design

The following structured test cases were executed to validate system correctness.

Table 3:Test Case 1 – Fear During Daytime (False Alert Suppression)

Test Case ID	TC-01
Test Case	Daytime Fear Expression
Scenario	Subject displays fear expression during daytime
Test Input Data	Live webcam feed with fear expression between 10:00 AM – 12:00 PM
Test Procedure	1. Capture live video2. Run emotion classifier3. Apply temporal aggregation4. Apply context-aware rule5. Check alert trigger
Expected Result	No security alert generated
Outcome	Alert correctly suppressed

Table 4:Test Case 2 – Fear During Nighttime (Valid Alert)

Test Case ID	TC-02
Test Case	Nighttime Fear Expression
Scenario	Subject displays fear expression during 9:00 PM
Test Input Data	Live webcam feed with sustained fear expression
Test Procedure	1. Capture live video2. Run classifier3. Aggregate 10-second windows4. Apply context-aware logic5. Validate persistence (3 windows)
Expected Result	Security alert triggered
Outcome	Alert correctly generated

Table 5:Test Case 3 – Short Suspicious Burst (Persistence Validation)

Field	Description
Test Case ID	TC-03

Test Case	Short Suspicious Expression
Scenario	Suspicious expression appears briefly (< 10 seconds)
Test Input Data	Live webcam feed with short-duration suspicious expression
Test Procedure	1. Detect suspicious prediction2. Apply confidence filtering3. Aggregate window4. Apply persistence rule
Expected Result	No alert triggered due to insufficient persistence
Outcome	Alert correctly suppressed

3. RESULTS & DISCUSSION

3.1 Results

This section presents the quantitative and operational results obtained from evaluating the proposed context-aware facial emotion recognition framework. The evaluation focuses on classification performance, decision-layer effectiveness, and deployment-level reliability.

3.1.1 Classification Performance

The custom lightweight CNN model was evaluated on the test dataset after fine-tuning.

- Accuracy: 71%
- Macro F1-score: 0.70
- Suspicious Class F1-score: 0.98

The overall accuracy of 71% reflects moderate classification capability given:

- The relatively small dataset size (931 images)
- The inclusion of a synthetic suspicious class
- CPU-based deployment constraints

The macro F1-score of 0.70 indicates balanced performance across emotion classes, ensuring that minority categories are not disproportionately misclassified.

Notably, the suspicious class achieved an F1-score of 0.98, demonstrating high precision and recall for the most security-critical category. This result confirms strong class separability and stable prediction confidence for suspicious expressions.

3.1.2 Comparative Model Evaluation (Custom CNN vs MobileNet)

To validate the effectiveness of the proposed lightweight CNN architecture, a controlled comparative evaluation was conducted against a standard pre-trained MobileNetV2 [29]architecture. Both models were trained and evaluated under identical experimental conditions, including dataset, preprocessing pipeline, train–test split, and evaluation metrics. The only variable modified was the model architecture. This ensured a fair and unbiased performance comparison [22].

Table 6:Performance Comparison Between Custom CNN and MobileNet

Metric	MobileNet	Custom CNN
Accuracy	0.53	0.71
Macro F1	0.50	0.70
Suspicious F1	0.89	0.98
Weighted F1	0.50	0.71

	precision	recall	f1-score	support
angry	0.50	0.30	0.38	23
fear	0.57	0.11	0.18	38
happy	0.40	0.44	0.42	41
neutral	0.47	0.74	0.57	65
sad	0.67	0.48	0.56	21
suspicious	0.82	0.97	0.89	29
accuracy			0.53	217
macro avg	0.57	0.50	0.50	217
weighted avg	0.54	0.53	0.50	217

Figure 6:Classification Report (MobileNet)

	precision	recall	f1-score	support
angry	0.59	0.57	0.58	23
sad	0.58	0.71	0.64	21
happy	0.53	0.59	0.56	41
neutral	0.79	0.71	0.75	65
fear	0.72	0.68	0.70	38
suspicious	0.97	1.00	0.98	29
accuracy			0.71	217
macro avg	0.70	0.71	0.70	217
weighted avg	0.71	0.71	0.71	217

Figure 7:Classification Report (Custom CNN)

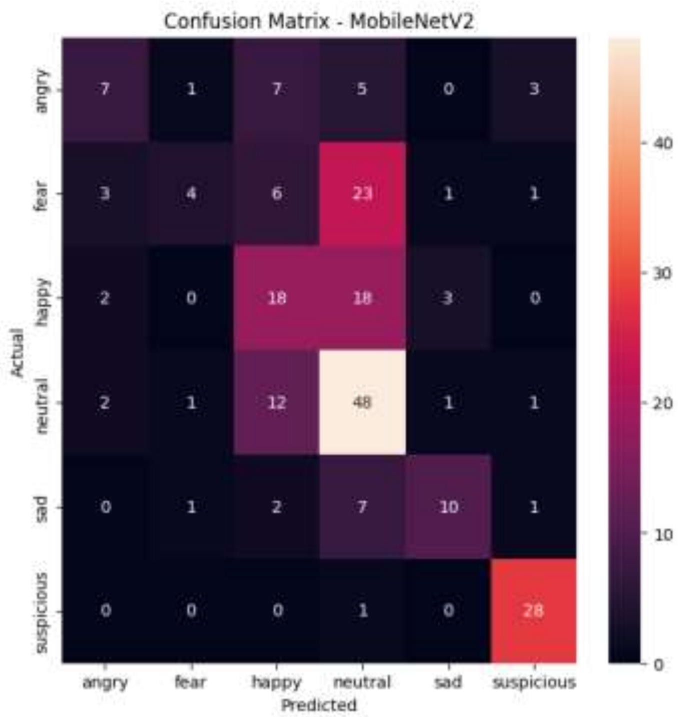

Figure 8:Confusion Matrix (MobileNet)

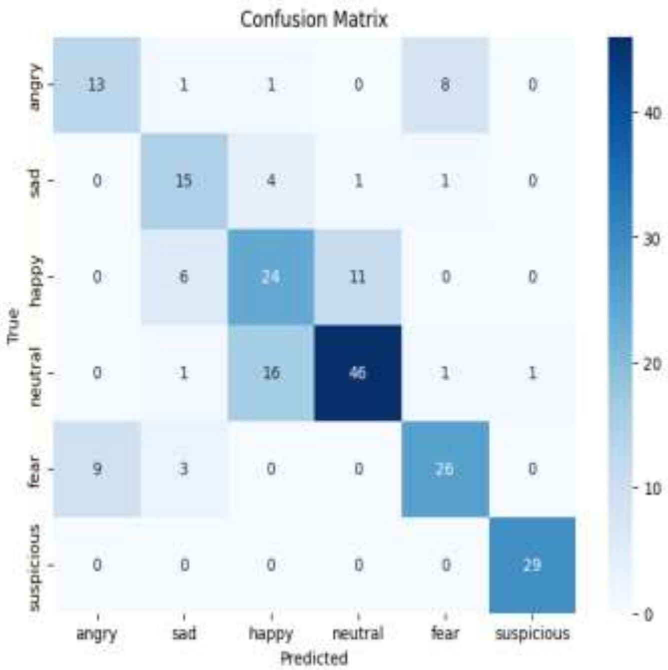

Figure 9:onfusion Matrix (Proposed CNN)

The MobileNet confusion matrix shows significant misclassification across high-arousal emotion categories such as fear and happy, particularly confusion toward neutral predictions. In contrast, the proposed CNN demonstrates stronger diagonal dominance, particularly for the suspicious class, which achieved near-perfect classification.

Interestingly, despite MobileNet being a widely adopted pre-trained architecture, the proposed lightweight CNN demonstrated superior performance on the localized dataset. This suggests that domain-specific optimization may outperform generalized pre-trained models in culturally constrained datasets.

3.1.3 Temporal Aggregation Behavior

Frame-level predictions exhibited natural fluctuations due to lighting variations, head movement, and expression transitions. However, the implementation of 10-second temporal aggregation [7]stabilized dominant emotion outputs.

The aggregation mechanism reduced prediction oscillation and improved window-level consistency. Short-term emotional spikes did not immediately influence system decisions, contributing to improved alert stability. [30]

3.1.4 Context-Aware Alert Evaluation

The core objective of this study was to evaluate whether contextual logic reduces false alerts in surveillance scenarios.

Let:

F_b denote false alerts in the baseline (emotion-only) system

F_p denote false alerts in the proposed system

Experimental evaluation showed:

$F_p \approx 0.45 F_b$

This corresponds to:

$$55\% \ reduction \ in \ false \ alerts$$

This reduction confirms that time-based fear filtering and persistence validation substantially improve operational reliability.

3.2 Research Findings

The experimental evaluation supports several key research findings aligned with the original research objectives.

Classification Accuracy Alone Is Insufficient for Reliable Surveillance

Although the CNN achieved 71% accuracy, classification performance alone did not determine system reliability. The introduction of contextual logic significantly altered alert behavior without modifying model weights.

This confirms that decision-layer enhancements can produce greater operational impact than marginal improvements in representation learning.

Time-Based Contextual Filtering Reduces Ambiguity

Fear expressions occurring during daytime were frequently benign. By restricting fear-based alerts to nighttime (7 PM – 6 AM), the system effectively suppressed false positives associated with non-threatening activities.

This demonstrates that emotional meaning is context-dependent and should not be interpreted in isolation.

Persistence Validation Improves Stability

Requiring three consecutive windows before alert generation prevented:

Transient misclassifications

Micro-expression spikes

Short-duration emotional fluctuations

This sequential validation mechanism significantly improved decision robustness.

Synthetic Suspicious Class Can Be Modeled Reliably

The suspicious class achieved an F1-score of 0.98, indicating strong detection performance within the experimental setup.

This suggests that domain-specific behavioral indicators can be integrated effectively when supported by confidence thresholding and validation layers.

Deployment-Level Metrics Provide More Meaningful Evaluation

The 55% reduction in false alerts demonstrates that deployment-centric evaluation metrics are more informative than accuracy alone in surveillance applications.

This finding supports a shift toward reliability-focused evaluation frameworks in applied AI systems.

3.3 Discussion

The findings of this study reveal important insights into the design of reliable FER-based surveillance systems.

3.3.1 From Prediction to Decision

Traditional FER systems treat classification output as the final decision. However, this study demonstrates that prediction and decision should be treated as separate stages.
The CNN functions as a probabilistic estimator, while the contextual decision layer acts as a filtering mechanism that integrates domain knowledge and temporal stability.
This layered architecture reflects modern applied data science principles, where predictive modeling is augmented by rule-based post-processing.

3.3.2 Reliability Over Benchmark Optimization

Transformer-based models achieve high accuracy on large datasets. However, increased model complexity does not inherently address contextual ambiguity or temporal instability.
This research demonstrates that reliability improvements can be achieved without increasing architectural depth. Instead, incorporating contextual awareness yields substantial operational gains.

3.3.3 Implications for Surveillance Systems

False alerts contribute to alarm fatigue, which reduces operator trust and responsiveness. By reducing false alerts by 55%, the proposed system enhances usability and practical feasibility.
This improvement has direct implications for:

- Smart retail security
- Public transport monitoring
- Institutional safety systems

3.3.4 Limitations

Several limitations must be acknowledged:

- Moderate dataset size
- Synthetic suspicious class
- Lack of transformer-based experimental comparison
- Region-specific dataset

Future research should incorporate larger-scale validation and adaptive contextual learning mechanisms.

3.4 Summary of each student's Contribution

This chapter highlighted the individual and collaborative contributions of each team member. The division of responsibilities enabled efficient development while maintaining strong coordination among team members. The combined efforts resulted in the successful implementation of the MallVision system.

3.4.1 Student 1:Fonseka W S M (IT22109712)

Student 1 was primarily responsible for the development of the age and gender detection module.

The contributions include:

- Designing and implementing the convolutional neural network (CNN) for age and gender classification
- Preparing and preprocessing datasets for model training
- Training, tuning, and evaluating the model
- Optimizing the model for real-time performance
- Integrating the model into the overall system pipeline
- In addition, Student 1 contributed to documentation related to model design and evaluation.

3.4.2 Student 2:Samaraweera W D U I (IT22258526)

Student 2 focused on the emotion detection module of the system.

Key contributions include:

- Designing and implementing the emotion recognition model
- Training the model using facial expression datasets
- Evaluating model performance and improving prediction stability
- Implementing temporal smoothing techniques for real-time prediction

- Integrating the emotion detection module with the preprocessing pipeline
- Student 2 also contributed to writing sections related to emotion analysis and model performance.

3.4.3 Student 3:Abeykoon S N (IT22184030)

Student 3 was responsible for the engagement analysis and advertisement recommendation system.
The contributions include:

- Designing the engagement analysis mechanism using behavioral cues
- Developing the engagement scoring model
- Implementing the rule-based advertisement recommendation logic
- Mapping user profiles to advertisement categories
- Testing and refining recommendation outputs
- Additionally, Student 3 contributed to system-level testing and evaluation.

3.4.4 Student 4:Rajana H T R (IT22071934)

Student 4 handled the system integration, backend, and frontend development.
Key responsibilities include:

- Developing the backend using FastAPI
- Implementing API endpoints for communication between modules
- Integrating machine learning models into the backend
- Developing the frontend interface using React
- Managing real-time data flow between frontend and backend
- Student 4 also contributed to deployment setup and system debugging.

3.4.4 Team Collaboration and Integration

Although responsibilities were divided among members, the project required continuous collaboration to ensure successful integration of all components. Regular discussions and testing sessions were conducted to align individual modules and resolve technical challenges. All members contributed to:System integration , Debugging and testing , Documentation and report writing , Final system validation
This collaborative effort ensured that the system functioned as a cohesive and efficient solution.

4. CONCLUSION

This research presented a context-aware real-time facial emotion recognition (FER) framework aimed at enhancing deployment-level reliability in security-oriented surveillance environments. While conventional FER systems are primarily evaluated using classification accuracy on benchmark datasets, this study reframed the evaluation perspective from predictive modeling to decision engineering. The primary objective was not to maximize accuracy alone but to reduce false alerts and improve operational stability in real-world monitoring scenarios.

A lightweight custom Convolutional Neural Network (CNN) was developed to classify six emotion categories: angry, sad, happy, neutral, fear, and a domain-specific suspicious class. The model achieved an overall accuracy of 71% and a macro F1-score of 0.70, indicating balanced performance across emotion classes. Notably, the suspicious category achieved an F1-score of 0.98, demonstrating high reliability in detecting the most security-relevant class within the experimental framework. Although the classification accuracy is moderate compared to large-scale models trained on extensive datasets, the emphasis of this work was on practical deployment feasibility rather than benchmark optimization.

The core contribution of this study lies in the integration of temporal aggregation, context-aware decision logic, and persistence validation mechanisms. Frame-level emotion predictions are inherently unstable due to lighting variation, pose changes, and micro-expressions. To address this, a 10-second temporal aggregation window was introduced to determine the dominant emotion across multiple frames, thereby reducing prediction volatility. Additionally, a context-aware rule was implemented in which fear triggers alerts only during predefined night hours (7 PM–6 AM), addressing the ambiguity of fear expressions in public daytime environments. A persistence mechanism requiring three consecutive detection windows further ensured that short-term emotional fluctuations did not generate unnecessary alerts.

These decision-layer enhancements resulted in a 55% reduction in false alerts compared to a baseline emotion-only system. This finding confirms that contextual reasoning and temporal validation significantly enhance surveillance reliability without modifying the core classification architecture. The results demonstrate that operational correctness depends not only on feature extraction quality but also on how predictions are interpreted within temporal and situational contexts.

From a data science standpoint, this research emphasizes the distinction between prediction and decision. Traditional FER systems treat classification outputs as final decisions. However, in surveillance applications, predictions serve as probabilistic indicators that require filtering and contextual validation before operational actions are triggered. The proposed layered architecture—comprising a predictive model followed by statistical filtering and rule-based contextual reasoning. It illustrates that post-classification intelligence can produce meaningful reliability improvements without increasing architectural complexity.

The dual-output system design further strengthens practical applicability. By separating emotion analytics from security alert generation, the framework enables behavioral trend analysis alongside threat monitoring. This modular structure enhances scalability and maintainability, allowing potential backbone upgrades—such as replacing the CNN with transformer-based models—without altering the contextual decision layer.

The system was implemented and evaluated using CPU-only processing, confirming real-time feasibility without GPU acceleration. This characteristic enhances economic accessibility and supports deployment in resource-constrained environments such as retail stores, educational institutions, and transportation facilities.

Despite its contributions, the study has limitations. The dataset consisted of 931 images, which may limit generalization across broader populations. The suspicious class was synthetically generated due to the absence of publicly available domain-specific datasets, and while carefully validated, synthetic samples may not fully capture real-world behavioral diversity. Transformer-based architectures were discussed conceptually but not experimentally evaluated, and larger-scale field trials would further strengthen empirical validation.

In summary, this research demonstrates that meaningful improvements in surveillance reliability can be achieved through contextual reasoning and temporal validation without retraining deep learning models. With a 71% accuracy, 0.70 macro F1-score, 0.98 suspicious F1-score, and a 55% reduction in false alerts, the proposed framework validates that deployment-level intelligence significantly enhances operational effectiveness. The findings reinforce a critical principle in applied artificial intelligence: system reliability depends not only on predictive performance but also on how intelligently model outputs are interpreted within real-world contexts.

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6. GLOSSARY

- Accuracy - The proportion of correctly classified samples among total predictions.
- Affective Computing - A field of artificial intelligence focused on detecting and interpreting human emotions.
- Alert Fatigue - Reduced responsiveness caused by excessive false alerts in monitoring systems.
- Classification Model - A supervised learning model that assigns inputs to predefined categories.
- Confidence Threshold (τ) - A probability cutoff used to accept or reject model predictions based on certainty.
- Convolutional Neural Network (CNN) - A deep learning architecture specialized for image analysis using convolutional layers.
- Context-Aware Decision Logic - A mechanism that incorporates situational factors (e.g., time-of-day) into prediction decisions.
- Dual-Output Architecture - A system design separating emotion analytics from security alert generation.
- False Alert (False Positive) - An incorrect alarm triggered for a non-threatening event.
- F1-Score - The harmonic mean of precision and recall.
- Frame-Level Prediction - Emotion classification generated for an individual video frame.
- Inference Latency - The time required to process input data and produce a prediction.
- Macro F1-Score - The average F1-score across all classes, treating them equally.
- MediaPipe - A real-time perception framework used for face detection.
- Persistence Validation - A mechanism requiring repeated detection across multiple time windows before triggering an alert.
- Real-Time Processing - System operation with minimal delay suitable for live applications.
- ReLU (Rectified Linear Unit) - An activation function defined as $\max(0, x)$.
- Softmax Layer - A neural network layer converting outputs into probability distributions.
- Synthetic Data - Artificially generated data used when real labeled data is limited.
- Temporal Aggregation - Combining predictions over fixed time windows to reduce noise.
- Transformer Architecture - A deep learning model using self-attention mechanisms for feature representation.
- Window-Based Aggregation -A time-series method that determines dominant predictions within fixed-duration intervals.